



How HTTPS Really Works

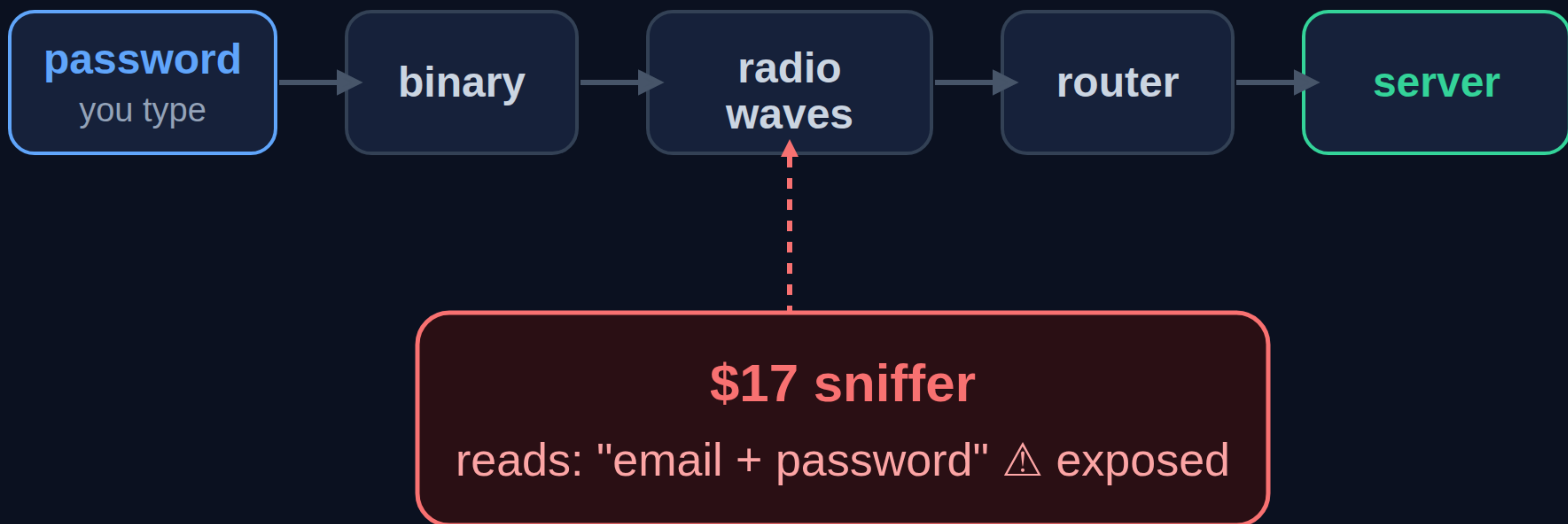
— *explained with diagrams* —

CAs prove who you're talking to

Asymmetric agree on a key, safely, once

Symmetric encrypt all the data, fast

1 Plain HTTP leaks everything



Browsers flag this as **"Not Secure."**

The fix: **encryption**, built in 3 layers

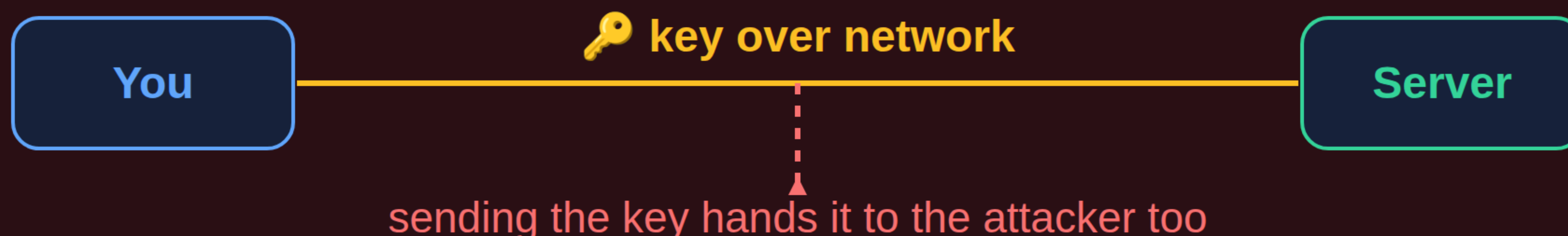
→ *symmetric · asymmetric · certificate authorities*

2 Symmetric one shared key



⚠ The flaw

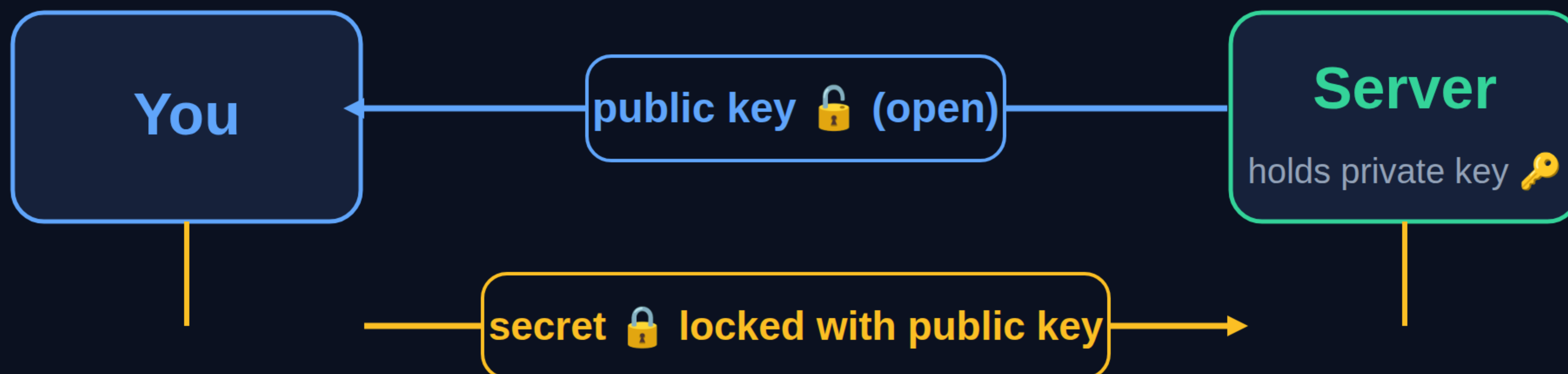
Both sides need the **same key** — so it has to be shared first.



3

Asymmetric

a key pair fixes the sharing problem



only the private key unlocks it — and it never leaves the server

✓ **Both now share a secret**

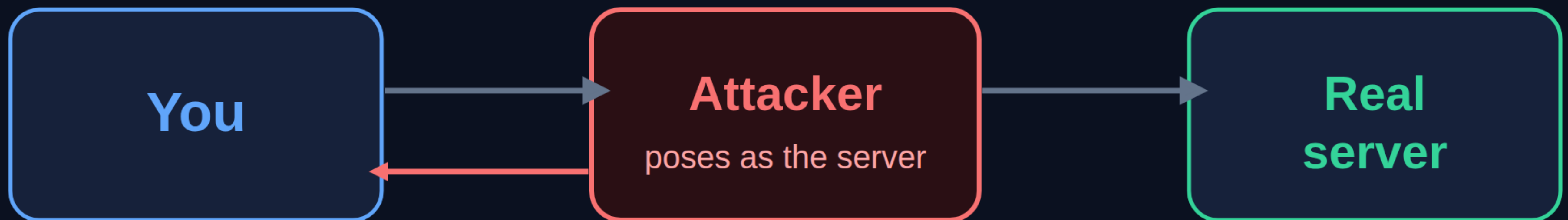
an eavesdropper can't open

...but one attack remains →

4

Man-in-the-middle

who really sent that public key?



attacker's OWN public key 🗝️

...and you can't tell the difference

The missing piece

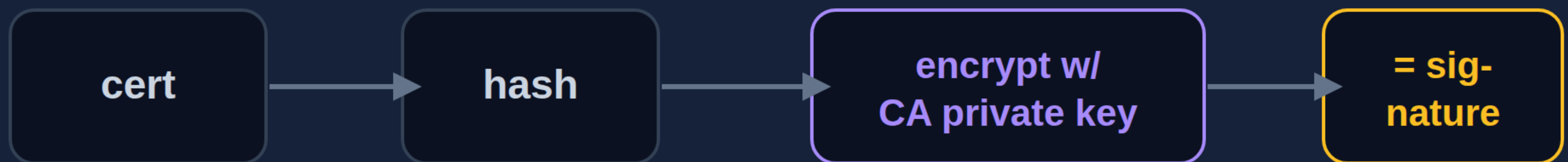
We must **prove** a public key really belongs
to the **real server**

5

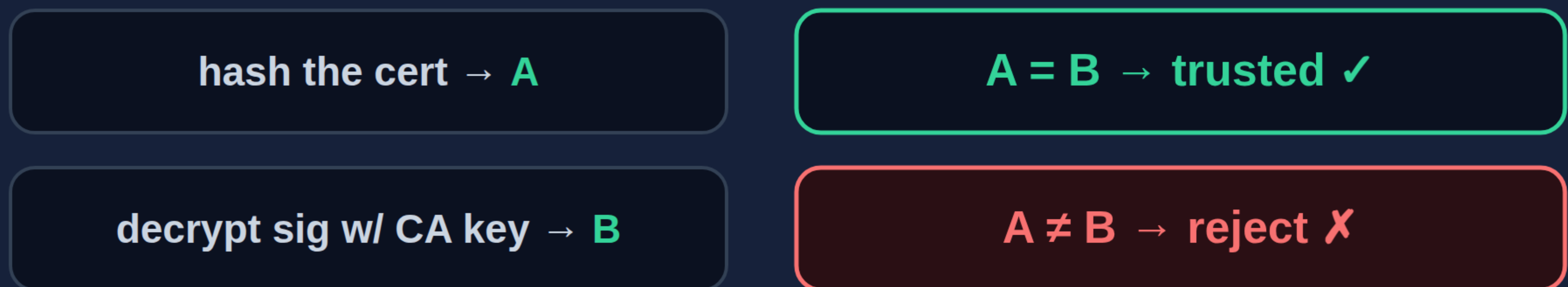
Certificate Authorities

proof of identity · DigiCert, Let's Encrypt

① CA signs (once)



② Browser checks



Proves **integrity + authenticity**

up a chain of trust to a trusted root

6

Session key

asymmetric hands off to symmetric

shared secret + randoms

the material both sides now hold

key derivation → session keys 🔑

"Finished"

confirms no tampering happened

switch to fast symmetric (AES)

for all data from here on

slow public-key math sets up the key · fast AES does the bulk work



TLS versions

and why 1.3 wins

1.0 / 1.1

weak · deprecated (2020–21)

1.2

common · RSA key exchange

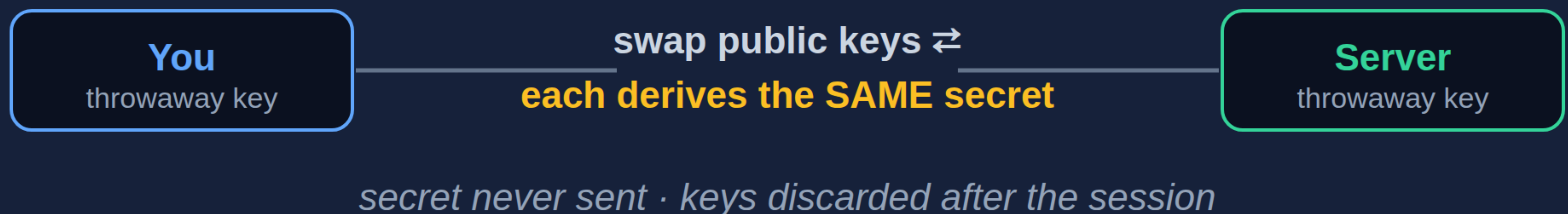
1.3

current · faster + safer ✓

RSA's risk

one long-lived key — leak it and **past traffic decrypts**

1.3 → ephemeral Diffie-Hellman



Leaked key can't unlock past traffic = forward secrecy

The big picture



CAs

certificate authorities

prove who you're talking to

Asymmetric

public / private key pair

agree on a key, safely, once

Symmetric

one shared session key (AES)

encrypt all the data, fast

Next time you see that padlock, you'll know what's behind it.

► [Full walkthrough \(video\)](#) — link in the post

#HTTPS #TLS #WebSecurity #Cryptography